

The CUHKSZ System for the IWSLT 2026 Low-Resource Speech-to-Text Task

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Abstract

This paper describes the CUHKSZ system for the IWSLT 2026 Low-Resource Speech-to-Text task. We propose Gradient-Driven Parameter Sharing (GDPS), a framework that analyzes inter-language gradient behaviors to automatically determine optimal language groupings and shared-private parameter ratios. Built upon SeamlessM4T-Medium, GDPS reduces negative transfer by specializing Layer 11 FFN2 while maintaining shared encoder representations across languages. Additionally, we incorporate curriculum distillation with progressive pseudo-label mixing and test-time reranking combining prior-BLEU weighting and self-consistency scoring. Evaluation on eight low-resource languages (bem, ckb, gle, hau, ibo, yor, aeb, est) demonstrates strongest gains on bem (+2.07 BLEU), hau (+1.50), and ibo (+0.38) compared to unified fine-tuning, while ckb and yor benefit more from prior-based reranking at inference.

1 Introduction

The IWSLT 2026 Low-Resource Speech-to-Text (S2TT) track focuses on multilingual speech translation for languages with limited training data. This task presents significant challenges due to the scarcity of parallel speech-text corpora and the linguistic diversity across target languages. The competition provides two evaluation settings: a constrained 6-language set (bem, ckb, gle, hau, ibo, yor) and an unconstrained 8-language set with additional languages (aeb, est).

Massively multilingual speech-to-text models such as SeamlessM4T (SEAMLESS Communication Team, 2025; Barrault et al., 2023b), Seamless v2 (Barrault et al., 2023a), and MMS (Pratap

et al., 2024) have shown strong cross-lingual transfer capabilities by pretraining on large multilingual datasets. However, when fine-tuning for multiple low-resource languages simultaneously, each language induces language-specific gradient signals that reflect distinct phonological and linguistic properties. These signals often conflict during back-propagation: parameter updates beneficial for one language may degrade performance on another, a problem known as gradient conflict. This occurs because languages with limited data provide noisy training signals that pull optimization in divergent directions, leading to performance degradation particularly for underrepresented languages (Yu et al., 2020). This phenomenon, often termed “negative transfer,” remains a fundamental challenge in multilingual learning.

Gradient conflict is recognized as a primary bottleneck in multilingual learning. Existing solutions include: (1) optimization-based approaches such as PCGrad (Yu et al., 2020) and GradOPS (Zhu et al., 2025) that stabilize training via orthogonal gradient projections; (2) architectural strategies such as shared-private representations (Bousmalis et al., 2016) and task disentangling (Yang et al., 2026) that seek to decouple interfering parameters; and (3) mixture-of-experts approaches that route tokens to specialized modules.

To address these challenges, we propose a Gradient-Driven Parameter Sharing (GDPS) framework (Sun and Nakamura, 2026) that analyzes inter-language gradient behaviors to automatically determine optimal language groupings and shared-private parameter ratios. Our key insight is that the FFN2 layer (feed-forward sublayer) in Conformer encoders exhibits high parameter density and plays a crucial role in non-linear feature transformation, making it more malleable for language-specific features than attention modules. We specialize only the Layer 11 FFN2 based on gradient analysis findings, minimizing architectural changes while maxi-

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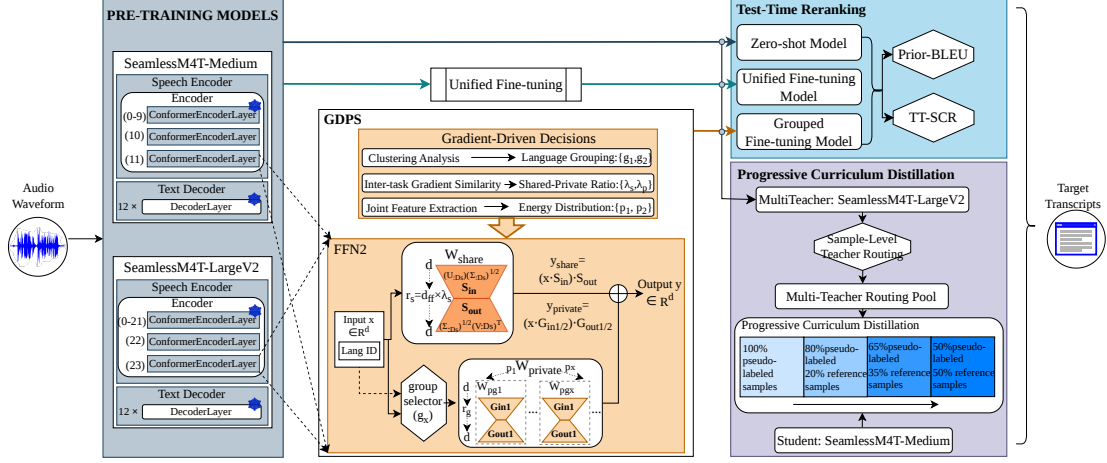


Figure 1: GDPS system overview with four main stages: (1) backbone model pre-training; (2) gradient conflict analysis via three methods to identify language groups and layer bottlenecks; (3) targeted FFN2 specialization with shared-private decomposition; (4) curriculum distillation training and test-time reranking.

mizing language-specific adaptation.

Our system builds upon SeamlessM4T-Medium and incorporates curriculum distillation using SeamlessM4T-v2-Large as teacher to bridge the gap between pretrained knowledge and low-resource fine-tuning. At inference time, we employ dual-strategy test-time reranking combining prior-BLEU weighting and test-time self-consistency reranking (TT-SCR) to select the best translation from multiple heterogeneous checkpoints.

This paper describes our system architecture, training pipeline, and experimental setup for the IWSLT 2026 evaluation. Our main contributions are:

- An automated gradient-driven framework that translates gradient statistics into optimal GDPS configurations for 8 languages
- A curriculum distillation protocol that stabilizes training through progressive data incorporation
- A dual-strategy test-time reranking framework for robust inference across heterogeneous checkpoints
- Comprehensive evaluation on 8 languages demonstrating where GDPS yields gains and where UFT remains stronger

2 System Description

2.1 Pre-trained Model

We adopt SeamlessM4T-Medium (Barrault et al., 2023b) as our backbone model, which is a mas-

sively multilingual speech-to-text model supporting up to 100 languages. The model consists of a speech encoder (12 Conformer layers) and a text decoder (12 layers), with approximately 1.2B parameters.

2.2 Data Processing

We conduct experiments on the IWSLT 2026 Low-resource Speech-to-Text track, covering two language sets:

- **8-Language Set (Unconstrained):** Tunisian Arabic (aeb), Bemba (bem), Central Kurdish (ckb), Estonian (est), Irish (gle), Hausa (hau), Yoruba (yor), Igbo (ibo).
- **6-Language Constraint Set (IWSLT 2026 Official):** bem, ckb, gle, hau, ibo, yor. This is the official evaluation set for the competition.

Dataset statistics are summarized in Table 1.

Table 1: Dataset summary. The 6-language subset corresponds to the IWSLT 2026 official constraint set.

Language	Task	Amount	Sources
Tunisian (aeb)	2-way ST	20k lines	IWSLT2022 (Anastasopoulos et al., 2022)
Bemba (bem)	2-way ST	20k lines	BIG-C (Sikasote et al., 2023)
Central Kurdish (ckb)	2-way ST	9k lines	IWSLT 2026
Estonian (est)	2-way ST	20k lines	LoResMT (Sildam et al., 2024)
Irish (gle)	2-way ST	10k lines	IWSLT2023 (Agarwal et al., 2023)
Hausa (hau)	2-way ST	15k lines	IWSLT 2026
Yoruba (yor)	2-way ST	15k lines	IWSLT 2026
Igbo (ibo)	2-way ST	14k lines	IWSLT 2026

All speech inputs are preprocessed at 16kHz following IWSLT 2026 protocols (lowercase, no punctuation).

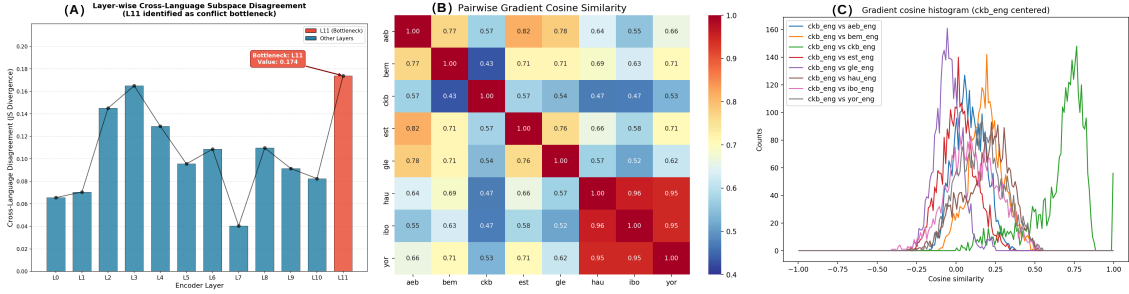


Figure 2: Gradient conflict analysis identifying bottlenecks and language groupings: (A) Layer-wise subspace disagreement reveals Layer 11 as the primary conflict point; (B) Pairwise gradient similarity shows ckb as a clear outlier with distinct cross-lingual patterns; (C) Individual language gradient concentration (Gini coefficient) demonstrates task-specificity variance across languages.

2.3 Training Strategy

Our training pipeline consists of three main stages:

2.3.1 Unified Fine-tuning Baseline

We first establish a unified fine-tuning baseline where all languages are jointly fine-tuned with uniform parameter sharing. This captures cross-lingual transfer through shared weights and serves as the starting point for GDPS specialization.

Key training hyperparameters are summarized in Table 2.

Table 2: Training configuration summary.

Parameter	Value
Base Learning Rate (α)	4×10^{-5}
Group-Adjusted Learning Rate (α_g)	1×10^{-4}
Batch Size (B)	4
Dropout (p_d)	0.05
Weight Decay (λ)	0.05
Warmup Steps (t_w)	2000
Seed (s)	2343
Optimizer	AdamW
Precision	FP16
Hardware	NVIDIA A100

Training uses mixed-precision (FP16) with AdamW optimizer on NVIDIA A100 GPUs.

2.3.2 Gradient-Driven Parameter Sharing (GDPS)

GDPS (Sun and Nakamura, 2026) automatically discovers the optimal parameter sharing configuration by analyzing inter-language gradient behaviors. The framework consists of three components:

Method A - Language Grouping via Clustering: We compute pairwise gradient cosine similarity between languages at the layer level. Let $\bar{g}_i \in \mathbb{R}^d$ denote the averaged gradient vector of

language i across all samples at a given layer. The pairwise cosine similarity is:

$$s_{i,j} = \frac{\bar{g}_i \cdot \bar{g}_j}{\|\bar{g}_i\| \|\bar{g}_j\|} \quad (1)$$

We convert similarity to distance $d_{i,j} = 1 - s_{i,j}$ and apply hierarchical clustering with Ward linkage to partition languages into groups sharing parameters. As illustrated in Figure 2(B), the 8×8 pairwise similarity heatmap reveals distinct language clusters.

Method B - Self versus Cross Gradient Similarity: To identify which languages are pulling parameters in conflicting directions, we compute intra-task and inter-task gradient similarity. Let $\mathbf{g}_{t,i}$ denote the gradient vector for language i at a sample t . We compute:

$$S_{\text{self}} = \mathbb{E}_{i \neq j} \mathbb{E}_t \cos(\mathbf{g}_{t,i}, \mathbf{g}_{t,j}) \quad (2)$$

$$S_{\text{cross}} = \mathbb{E}_{i,j} \mathbb{E}_{t \neq t'} \cos(\mathbf{g}_{t,i}, \mathbf{g}_{t',j}) \quad (3)$$

The conflict strength is $\delta = S_{\text{self}} - S_{\text{cross}}$: high values indicate that language pairs agree strongly within batches but diverge across batches, suggesting batch-level gradient conflicts. This score maps to a shared ratio:

$$\text{SharedRatio} = \begin{cases} 0.75 & \delta < 0.05 \\ 0.50 & 0.05 \leq \delta < 0.15 \\ 0.25 & \delta \geq 0.15 \end{cases} \quad (4)$$

Figure 2(C) presents the sample-level analysis, where ckb exhibits notably higher self-similarity (0.509) compared to cross-language pairs (~ 0.21), confirming its outlier status.

Method C - Joint SVD and Regularized CCA: Let $\mathbf{G}_i \in \mathbb{R}^{m \times d}$ be the gradient matrix of language i (stacking sample-level gradients across m samples with d dimensions).

We concatenate all language gradient matrices as $\mathbf{G}_{\text{concat}} = [\mathbf{G}_1; \dots; \mathbf{G}_n] = \mathbf{U}\Sigma\mathbf{V}^\top$ via SVD. Ridge-regularized CCA maximizes cross-covariance between language pairs. The energy concentration for language i is $p_i = E_i / \sum_{l=1}^n E_l$, where $E_i = \sum_{j=1}^k \|\mathbf{G}_i \mathbf{v}_j\|^2$ measures the cumulative projection energy onto the top- k singular vectors.

Layer Selection: As shown in Figure 2(A), cross-language subspace disagreement computed via CCA across all 12 encoder layers reveals that Layer 11 FFN2 exhibits the highest disagreement (0.174), confirming it as the primary gradient conflict bottleneck. This layer is selected for FFN2 specialization.

Language Grouping Results: Hierarchical clustering with Ward linkage on the 8×8 distance matrix yields three clusters at $K=3$ cut:

- Group 0: ckb (Central Kurdish) - isolated cluster with weak connectivity (silhouette 0.33), indicating its gradient profile differs substantially from all other languages
- Group 1: hau, ibo, yor (Niger-Congo family) - tight cluster (silhouette ~ 0.55) reflecting high mutual gradient similarity
- Group 2: aeb, bem, est, gle (European/Afro-Asiatic languages) - cohesive cluster (silhouette ~ 0.48) with moderate shared gradients

Shared-Private Ratio Results: The conflict score $\delta \approx 0.08$ falls in the medium conflict range ($0.05 \leq \delta < 0.15$), yielding a 50% shared ratio. However, ckb exhibits $\delta > 0.15$, indicating it would benefit from a lower shared ratio.

Energy Distribution: Joint SVD shows high gradient concentration (Gini coefficients 0.74–0.85), with gle having the highest concentration (0.85) and ckb the lowest (0.74), reflecting ckb’s isolated gradient profile.

Figure 2 presents the complete gradient conflict analysis visualization.

Architecture Instantiation: Based on the gradient analysis findings, we specialize Encoder Layer 11 FFN2 by decomposing it into shared and language-group-specific components. This decomposition allows languages with high mutual similarity (e.g., hau-ibo-yor) to share common representations while permitting language-specific adaptations for outliers (e.g., ckb).

The original FFN layer has input/output dimension $d_{\text{model}} = 1024$ and intermediate dimension

$d_{\text{ff}} = 4096$. To systematically extract shared structure from the pre-trained weights, we decompose the original weight matrix $\mathbf{W}_{\text{unified}} = \mathbf{W}_2\mathbf{W}_1$ (where $\mathbf{W}_1 \in \mathbb{R}^{d_{\text{ff}} \times d_{\text{model}}}$ and $\mathbf{W}_2 \in \mathbb{R}^{d_{\text{model}} \times d_{\text{ff}}}$) into a shared and private factorization.

We first compute the equivalent Gram matrix $\mathbf{W}_{\text{equiv}} = (\mathbf{W}_2\mathbf{W}_1)^\top \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}$ and apply SVD:

$$\mathbf{W}_{\text{equiv}} = \mathbf{U}\Sigma\mathbf{V}^\top \quad (5)$$

The intuition is that top- k singular vectors (for $k = d_{\text{model}} \times \alpha = 512$ with $\alpha = 0.5$ shared ratio) capture universal linguistic structure useful across all languages, while remaining components capture language-specific variance. The shared component is initialized from:

$$\mathbf{W}_{s,1} = \mathbf{U}_{:,1:k} \sqrt{\Sigma_{1:k,1:k}} \in \mathbb{R}^{d_{\text{model}} \times k} \quad (6)$$

$$\mathbf{W}_{s,2} = \sqrt{\Sigma_{1:k,1:k}} \mathbf{V}_{:,1:k}^\top \in \mathbb{R}^{k \times d_{\text{model}}} \quad (7)$$

These factors are then expanded to the shared FFN intermediate dimension $r_s = d_{\text{ff}} \times \alpha = 2048$:

$$\widehat{\mathbf{W}}_{s,1} \in \mathbb{R}^{d_{\text{model}} \times r_s}, \quad \widehat{\mathbf{W}}_{s,2} \in \mathbb{R}^{r_s \times d_{\text{model}}} \quad (8)$$

The remaining private capacity $d_{\text{pv}} = d_{\text{ff}} \times (1 - \alpha) = 2048$ is allocated to language-group-specific parameters to capture divergent linguistic patterns. Following the gradient analysis (Method C), we distribute this capacity across $N = 3$ language groups proportional to their energy concentration $\beta_g = p_g / \sum_l p_l$ (where p_g is the cumulative energy for group g), yielding per-group private dimension $r_g = d_{\text{pv}} \times \beta_g$. This energy-based allocation ensures that language groups exhibiting stronger task-specific gradients receive larger private capacity for specialization.

The final FFN2 layer combines shared and private components:

$$\text{FFN2}(x) = \text{SiLU}(x\widehat{\mathbf{W}}_{s,1})\widehat{\mathbf{W}}_{s,2} \oplus \bigoplus_{g=1}^N \text{SiLU}(x\mathbf{W}_{g,1})\mathbf{W}_{g,2} \quad (9)$$

where $\mathbf{W}_{g,1} \in \mathbb{R}^{d_{\text{model}} \times r_g}$, $\mathbf{W}_{g,2} \in \mathbb{R}^{r_g \times d_{\text{model}}}$, and $\oplus$ denotes concatenation along the feature dimension. Through this factorization, the model maintains universal representations (shared path) while allowing each language group to adapt its feature representations (private paths).

Table 3: BLEU/CHRF++ on 8-language set. ZS=Zero-shot, UFT=Unified Fine-tuning, GDPS=Our method. Avg = macro-average across languages.

Method	aeb	bem	ckb	est	gle	hau	ibo	yor	Avg
ZS-Med	3.58/17.48	0.87/15.58	7.19/22.97	30.44/53.74	0.07/6.01	0.50/11.49	0.74/10.38	7.50/27.44	6.36/20.64
ZS-LargeV2	4.02/18.21	0.81/13.72	19.23/40.96	29.29/50.63	0.06/8.48	0.59/11.68	0.97/11.11	12.06/33.86	8.25/23.58
UFT-Med	7.70/24.19	18.49/41.58	10.67/29.17	40.06/62.04	0.10/10.38	1.80/20.64	3.00/20.44	8.43/29.39	11.28/29.73
GDPS-Med	8.42/25.79	20.56/43.70	8.38/26.46	39.09/61.35	0.16/10.14	3.30/23.38	3.38/20.76	8.03/29.20	11.54/30.10
UFT-LV2 (ref.)	8.66/26.79	18.32/39.78	19.03/40.65	47.96/68.32	0.08/11.57	1.58/20.11	1.96/18.87	13.20/35.07	13.85/32.52
GDPS-LV2 (ref.)	8.54/26.31	18.27/40.67	17.35/38.59	47.04/67.29	0.07/10.33	1.82/20.31	1.98/18.81	11.91/34.00	13.00/31.96

Table 4: BLEU/CHRF++: 8L vs 7L (ckb removed). ckb=N/A in 7L setting. Avg = macro-average across languages.

Model	aeb	bem	ckb	est	gle	hau	ibo	yor	Avg
8L-UFT-Med	7.70/24.19	18.49/41.58	10.67/29.17	40.06/62.04	0.10/10.38	1.80/20.64	3.00/20.44	8.43/29.39	11.28/29.73
8L-GDPS-Med	8.42/25.79	20.56/43.70	8.38/26.46	39.09/61.35	0.16/10.14	3.30/23.38	3.38/20.76	8.03/29.20	11.42/32.60
8L-UFT-LV2	8.66/26.79	18.32/39.78	19.03/40.65	47.96/68.32	0.08/11.57	1.58/20.11	1.96/18.87	13.20/35.07	13.85/38.01
8L-GDPS-LV2	8.54/26.31	18.27/40.67	17.35/38.59	47.04/67.29	0.07/10.33	1.82/20.31	1.98/18.81	11.91/34.00	13.00/35.73
7L-UFT-Med	7.62/24.52	18.91/41.84	N/A	40.40/62.22	0.09/10.25	2.02/21.16	3.20/20.62	7.81/29.09	11.32/30.83
7L-GDPS-Med	8.61/26.07	20.53/42.64	N/A	39.39/61.43	0.07/9.74	3.56/23.69	3.34/20.52	7.94/28.48	11.32/30.93
7L-UFT-LV2	8.67/26.42	17.34/39.69	N/A	48.66/68.60	0.11/11.05	1.51/19.55	1.90/19.01	12.95/34.99	13.77/35.98
7L-GDPS-LV2	9.39/28.35	19.57/41.86	N/A	48.00/68.19	0.12/9.46	2.30/21.56	2.31/19.44	11.57/33.42	13.47/35.56

2.3.3 Curriculum Distillation

For enhanced training stability, we apply curriculum distillation using SeamlessM4T-v2-Large as the teacher model. The protocol consists of four stages: **Stage 0 (Distill)** uses teacher-generated pseudo-labels; **Stage 1 (mix20)** uses 20% real data + 80% pseudo labels; **Stage 2 (mix35)** uses 35% real data + 65% pseudo labels; **Stage 3 (mix50)** uses 50% real data + 50% pseudo labels.

2.4 Inference Strategy

At inference time, we have multiple heterogeneous checkpoints: SeamlessM4T-Medium zero-shot, Unified FT checkpoints, and GDPS checkpoints. No single checkpoint uniformly dominates all languages. We propose a test-time reranking framework with two complementary strategies.

Let $\mathcal{M}$ denote the set of available model checkpoints, each producing a candidate translation y_m^i for sample i .

Prior-BLEU Weighted Selection: We assign each model a prior reliability score R_m estimated from historical validation BLEU. For each sample, we select the model m^* that maximizes:

$$m^* = \operatorname{argmax}_{m \in \mathcal{M}} \left[\alpha \cdot \tilde{R}_m + \beta \cdot \frac{1}{|\mathcal{M}|-1} \sum_{n \neq m} \operatorname{chrF}(y_m^i, y_n^i) - \gamma \cdot P(y_m^i) \right] \quad (10)$$

where $\tilde{R}_m$ is the min-max normalized prior score, chrF measures agreement with other model hypotheses (MBR-style), and $P(\cdot)$ is a degeneration

penalty. The weights (α, β, γ) are tuned on a validation set.

The prior scores $\tilde{R}_m$ encode cross-lingual transfer efficiency: unified fine-tuned models score higher on languages with abundant in-domain data, while GDPS models score higher on languages exhibiting strong gradient conflict during training.

Test-Time Self-Consistency Reranking (TT-SCR): TT-SCR is a prior-free alternative that relies entirely on test-time signals:

$$\text{TT-SCR}(m, i) = \frac{1}{|\mathcal{M}|-1} \sum_{n \neq m} \operatorname{chrF}(y_m^i, y_n^i) - \lambda \cdot P(y_m^i) \quad (11)$$

The selection is $m^* = \operatorname{argmax}_m \text{TT-SCR}(m, i)$.

TT-SCR exploits the insight that models producing translations agreeing with many other models tend to be better calibrated. To identify challenging samples, we compute consensus disagreement as $\Delta_i = \max_{m,n} \operatorname{chrF}(y_m^i, y_n^i) - \min_{m,n} \operatorname{chrF}(y_m^i, y_n^i)$. High disagreement indicates model uncertainty, which flags hard samples.

3 Experiments

3.1 Evaluation Setup

Evaluation follows the IWSLT 2026 official protocol: standard lowercase BLEU (sacreBLEU) without punctuation. We report BLEU and $\operatorname{chrF}++$ (Popović, 2015) on the validation set as primary metrics.

Table 5: BLEU/CHRF++ on curriculum distillation and test-time reranking. Avg = macro-average across languages.

Stage/Strategy	aeb	bem	ckb	est	gle	hau	ibo	yor	Avg
GDPS (baseline)	8.42/25.79	20.56/43.70	8.38/26.46	39.09/61.35	0.16/10.14	3.30/23.38	3.38/20.76	8.03/29.20	11.54/30.10
Distill-only	6.92/22.00	17.27/39.74	3.50/15.78	36.76/58.88	0.10/8.13	1.28/16.03	2.32/16.48	6.19/25.48	8.19/23.07
mix20	7.01/22.59	17.92/40.78	4.05/16.93	37.74/59.82	0.08/7.56	1.09/15.96	1.97/15.61	5.57/24.58	8.85/24.98
mix35	7.41/23.27	18.93/41.98	3.84/16.92	38.82/60.73	0.09/7.90	1.30/16.80	2.42/16.23	6.50/25.94	9.66/26.22
mix50	7.60/23.31	19.23/41.73	4.50/17.87	38.42/60.40	0.07/7.97	1.52/17.47	2.44/17.07	6.30/25.66	10.26/26.44
Prior-BLEU	9.43/28.35	20.60/43.84	19.23/40.96	48.07/68.33	0.12/9.59	3.56/23.69	3.38/20.76	13.20/35.07	14.85/34.46
Prior-BLEU+Consensus	9.62/28.65	20.56/43.71	20.26/42.09	48.00/68.21	0.12/9.46	3.56/23.69	3.43/20.93	13.23/35.11	14.99/34.60
TT-SCR	8.10/25.89	18.70/43.58	10.24/29.72	40.47/62.54	0.09/10.92	2.41/22.26	3.26/21.22	8.64/30.58	11.49/30.84

Table 6: BLEU/COMET on 8-language set. Avg = macro-average across languages.

Method	aeb	bem	ckb	est	gle	hau	ibo	yor	Avg
ZS-Med	3.58/ 0.513	0.87/0.414	7.19/0.510	30.44/0.726	0.07/0.379	0.50/0.384	0.74/0.395	7.50/0.565	6.36/0.484
ZS-LargeV2	4.02/0.510	0.81/0.402	19.23/0.661	29.29/0.737	0.06/0.395	0.59/0.389	0.97/0.401	12.06/0.628	8.38/0.490
UFT-Med	7.70/0.554	18.49/0.688	10.67/0.553	40.06/0.758	0.10/0.409	1.80/0.492	3.00/0.496	8.43/0.596	11.28/0.556
GDPS-Med	8.42/0.567	20.56/0.707	8.38/0.527	39.09/0.753	0.16/0.419	3.30/0.524	3.38/0.498	8.03/0.590	11.54/0.573
UFT-LV2 (ref.)	8.66/0.571	18.32/0.682	19.03/0.653	47.96/0.789	0.08/0.442	1.58/0.498	1.96/0.486	13.20/0.650	13.85/0.584
GDPS-LV2 (ref.)	8.54/0.567	18.27/0.684	17.35/0.642	47.04/0.786	0.07/0.389	1.82/0.501	1.98/0.486	11.91/0.645	13.00/0.600

3.2 Main Results

Table 3 presents BLEU/chrF++ results for all 8 language pairs.

Key observations: GDPS-Med shows largest gains over UFT-Med on bem (+2.07 BLEU), hau (+1.50 BLEU), aeb (+0.72 BLEU), and ibo (+0.38 BLEU), while UFT-Med is stronger on ckb (10.67 vs 8.38), est (40.06 vs 39.09), and yor (8.43 vs 8.03). The ckb result is consistent with its outlier status in gradient analysis (highest self-similarity 0.509, lowest Gini coefficient 0.74), and Prior-BLEU reranking recovers ckb performance to 20.26 BLEU.

3.3 Ablation: 7-Language Robustness

Table 4 compares 8-language vs. 7-language (ckb removed) configurations. Removing ckb benefits most languages, confirming its high gradient conflict degrades performance.

3.4 Ablation: Curriculum Distillation and Test-Time Reranking

Table 5 presents ablation results on curriculum distillation stages and test-time reranking strategies.

Curriculum Distillation Analysis: Progressive curriculum distillation (Distill-only → mix20 → mix35 → mix50) yields consistent improvements across most languages. The mix50 stage achieves the best overall performance, with notable gains on aeb (6.92 → 7.60 BLEU), ckb (3.50 → 4.50 BLEU), and hau (1.28 → 1.52 BLEU). Estonian (est) maintains high performance throughout (36.76 → 38.42 BLEU), while Irish (gle) remains challenging with near-zero BLEU due to severe re-

source constraints.

Test-Time Reranking Analysis: Prior-BLEU reranking substantially outperforms both the GDPS baseline and TT-SCR on most languages. Notably, Prior-BLEU achieves 19.23 BLEU on ckb (vs 8.38 GDPS baseline) and 13.20 BLEU on yor (vs 8.03), demonstrating that leveraging historical BLEU priors effectively addresses ckb’s gradient conflict at inference. TT-SCR shows mixed results, with gains on some languages (yor: 8.64 vs 8.03 baseline) but degradation on others, suggesting that the prior-free approach may struggle when model consensus does not align with translation quality. Prior-BLEU+Consensus achieves comparable or slightly better results than Prior-BLEU alone on several languages, validating the complementary benefit of consensus scoring.

3.5 Additional Evaluation: COMET

Table 6 presents COMET scores for all model configurations. COMET is a neural evaluation metric that correlates most closely with human judgment (Rei et al., 2020).

Table 7 presents results on the official IWSLT 2026 6-language constraint set.

4 Conclusion

This paper presented the CUHKSZ system for the IWSLT 2026 Low-Resource Speech-to-Text task. Our key contribution is GDPS (Gradient-Driven Parameter Sharing), a framework that leverages gradient analysis to automatically identify language groupings and optimal shared-private parameter ratios. Unlike uniform fine-tuning, GDPS achieves

Table 7: BLEU/CHRF++ on 6-language set. C=Constrained, UC=Unconstrained.

Setting	bem	ckb	gle	hau	ibo	yor
C-UFT-Med	16.08/37.57	7.81/25.98	0.07/9.67	1.03/16.81	2.04/17.23	6.91/26.17
C-GDPS-Med	18.17/41.04	7.15/24.84	0.06/10.43	1.35/18.66	1.93/17.67	5.60/25.63
C-UFT-LV2 (ref.)	18.03/39.63	15.51/37.18	0.08/10.53	1.55/19.52	1.87/18.73	12.36/33.93
C-GDPS-LV2 (ref.)	17.89/38.97	15.29/36.57	0.10/11.52	1.52/19.34	1.64/17.82	10.91/32.74
UC-UFT-Med	18.49/41.58	10.67/29.17	0.10/10.38	1.80/20.64	3.00/20.44	8.43/29.39
UC-GDPS-Med	20.56/43.70	8.38/26.46	0.16/10.14	3.30/23.38	3.38/20.76	8.03/29.20

substantial gains on languages with strong gradient conflicts: bem (+2.07 BLEU), hau (+1.50 BLEU), and ibo (+0.38 BLEU) compared to UFT-Med. Conversely, languages with isolated gradient profiles (ckb, yor) benefit more from prior-aware test-time reranking at inference.

Our experimental findings highlight four key insights: (1) GDPS specialization recovers performance for conflict-prone languages while maintaining gains via Prior-BLEU reranking (ckb: 8.38→20.26, yor: 8.03→13.20); (2) languages with high gradient conflict (indicated by low silhouette scores) degrade multilingual training, and their removal benefits other languages; (3) expanded language diversity during unconstrained training amplifies GDPS gains compared to constrained settings, particularly for bem, hau, and ibo; and (4) curriculum distillation and test-time reranking strategies provide complementary improvements, with Prior-BLEU+Consensus achieving 14.99 average BLEU.

Future work includes automating cluster number selection through information-theoretic criteria and extending gradient analysis to decoder layers for joint encoder-decoder specialization.

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